
Machine Learning and LLM Agents for Microscopy

Classical supervised and unsupervised learning, spectral unmixing, and agentic microscope control



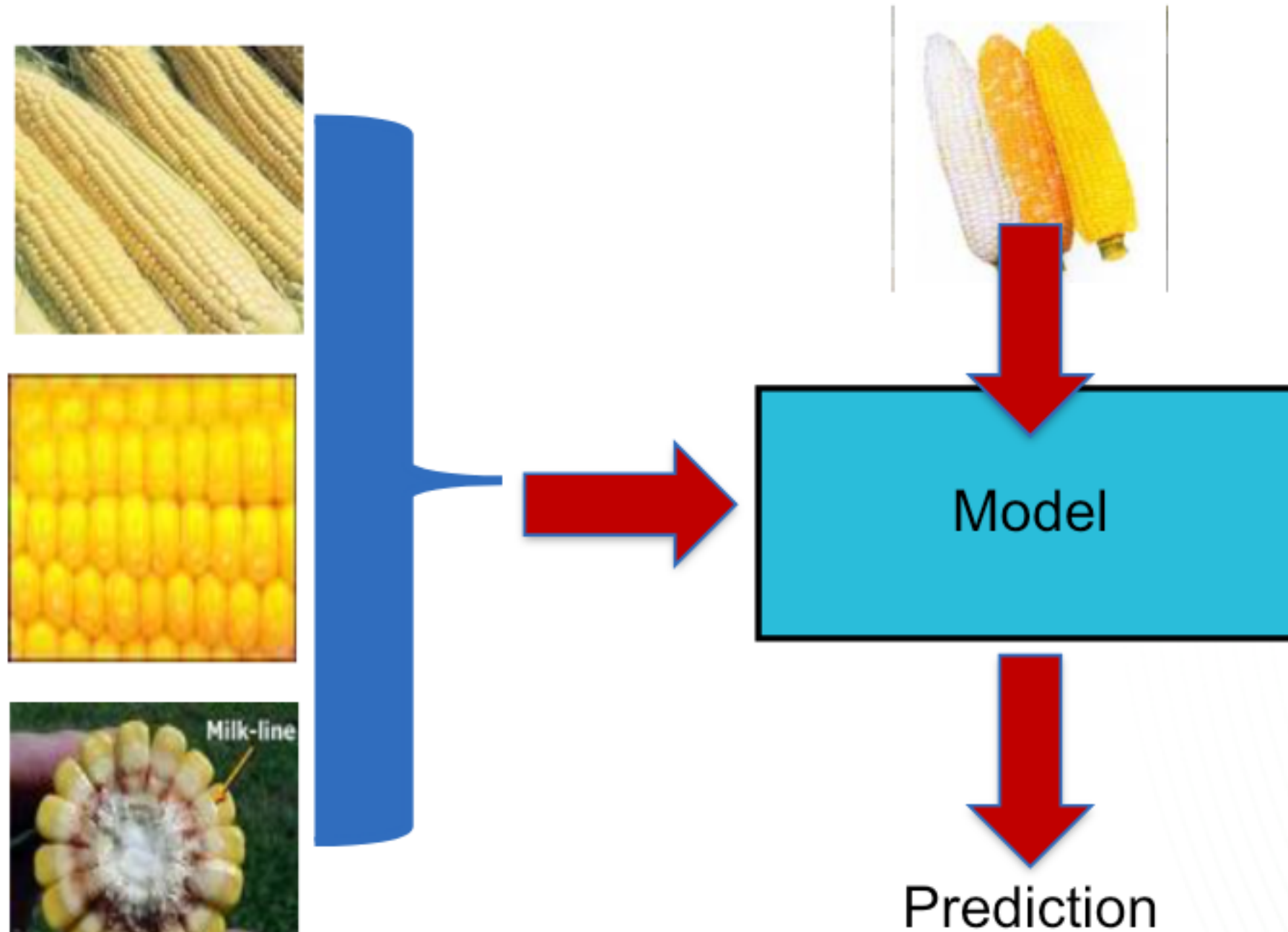
ML

Pre-processing

- Machine learning learns from data, without respect to what is an artifact and what is a real signal.
- As a result, care and appropriate pre-processing is required. Examples include step-changes in response, shifts of the signal in time, changes to intensity from artefacts (e.g. topographic), etc.
- Old adage: garbage in, garbage out.

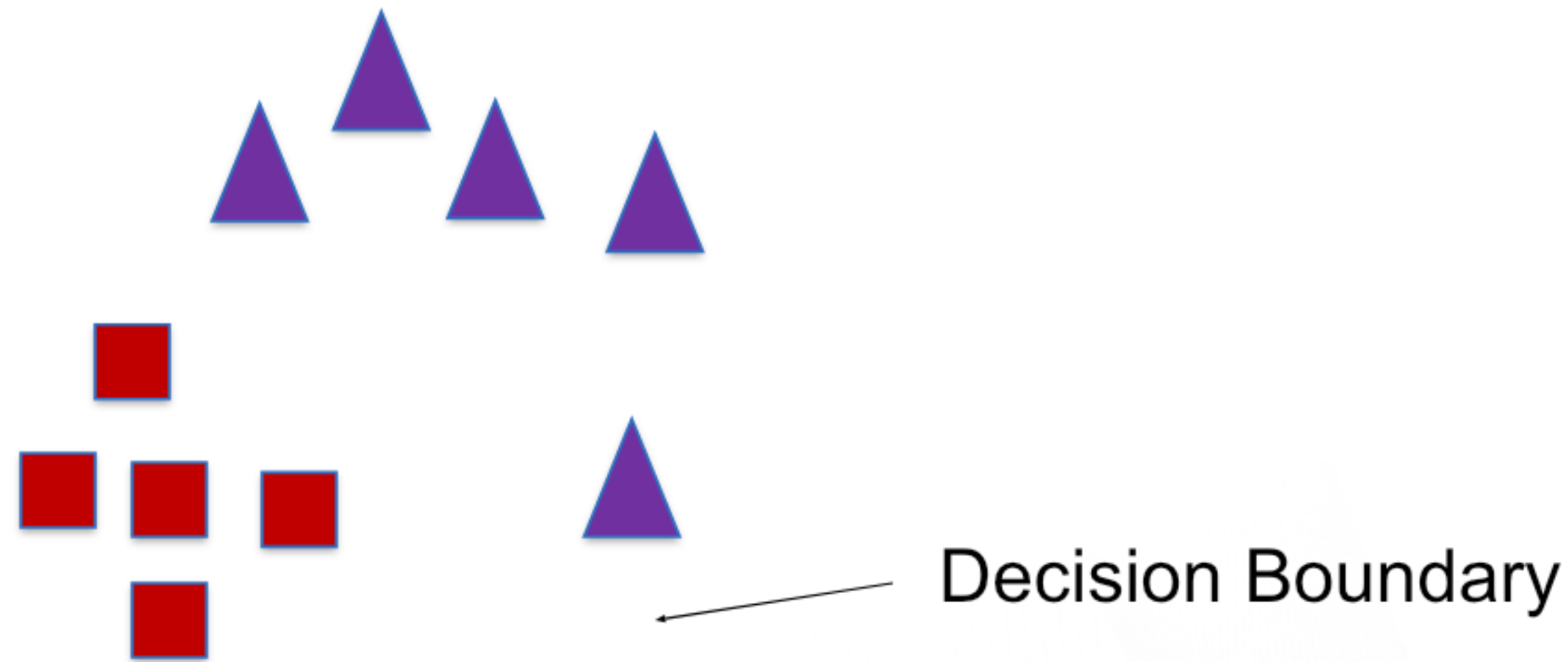
Supervised Classification

- Supervised classifiers can automatically classify data once the model has been trained on a 'training set'.



Supervised Classification

- Support Vector Machines are one common method, and is conceptually easy to understand.

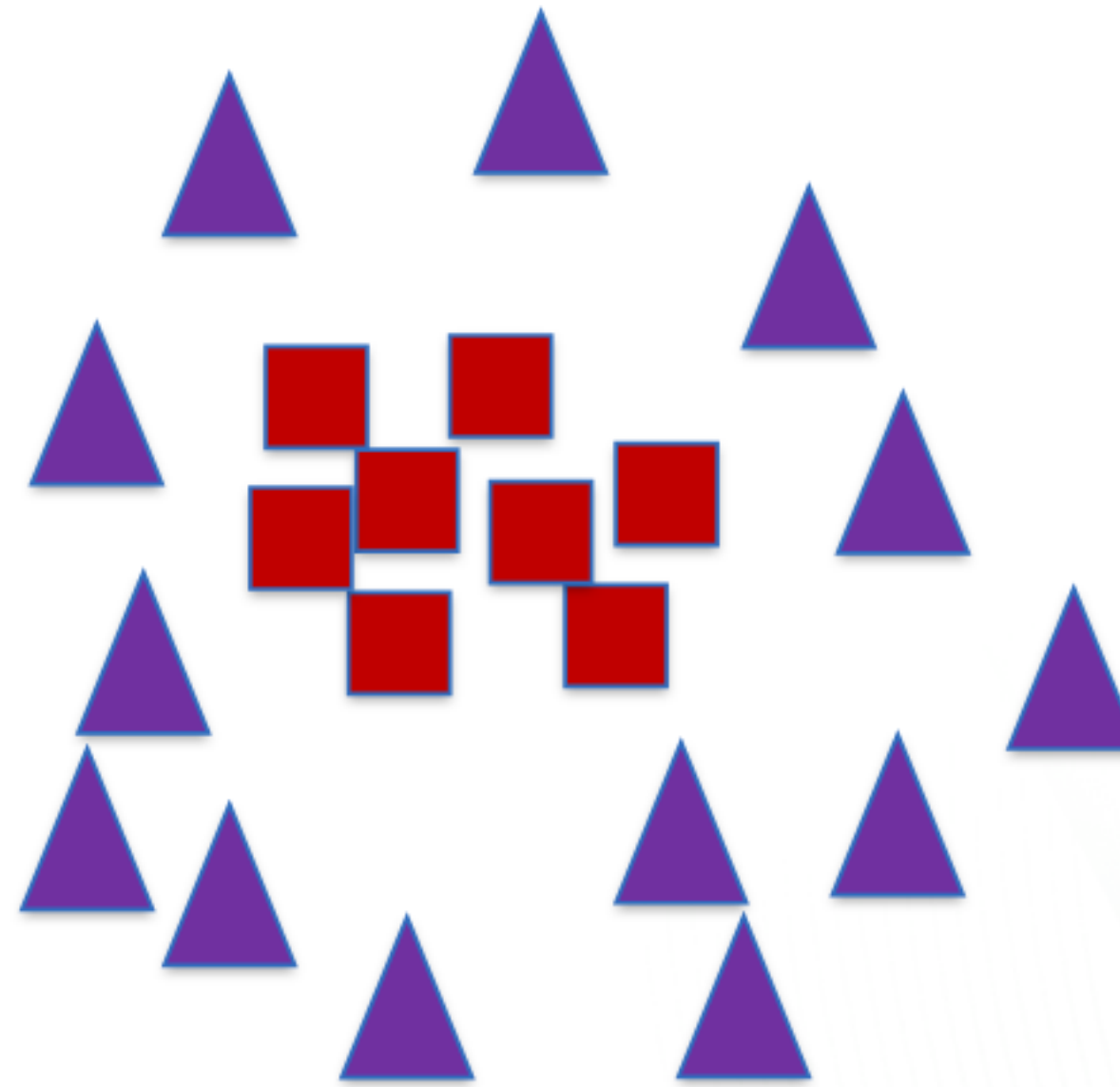


SVM are linear classifiers that fits lines (planes/hyperplanes in higher dimensions), such that the margin (separation between the line and the nearest training point) is maximized.

Supervised Classification

- But what to do when it cannot be linearly separated? E.g.,

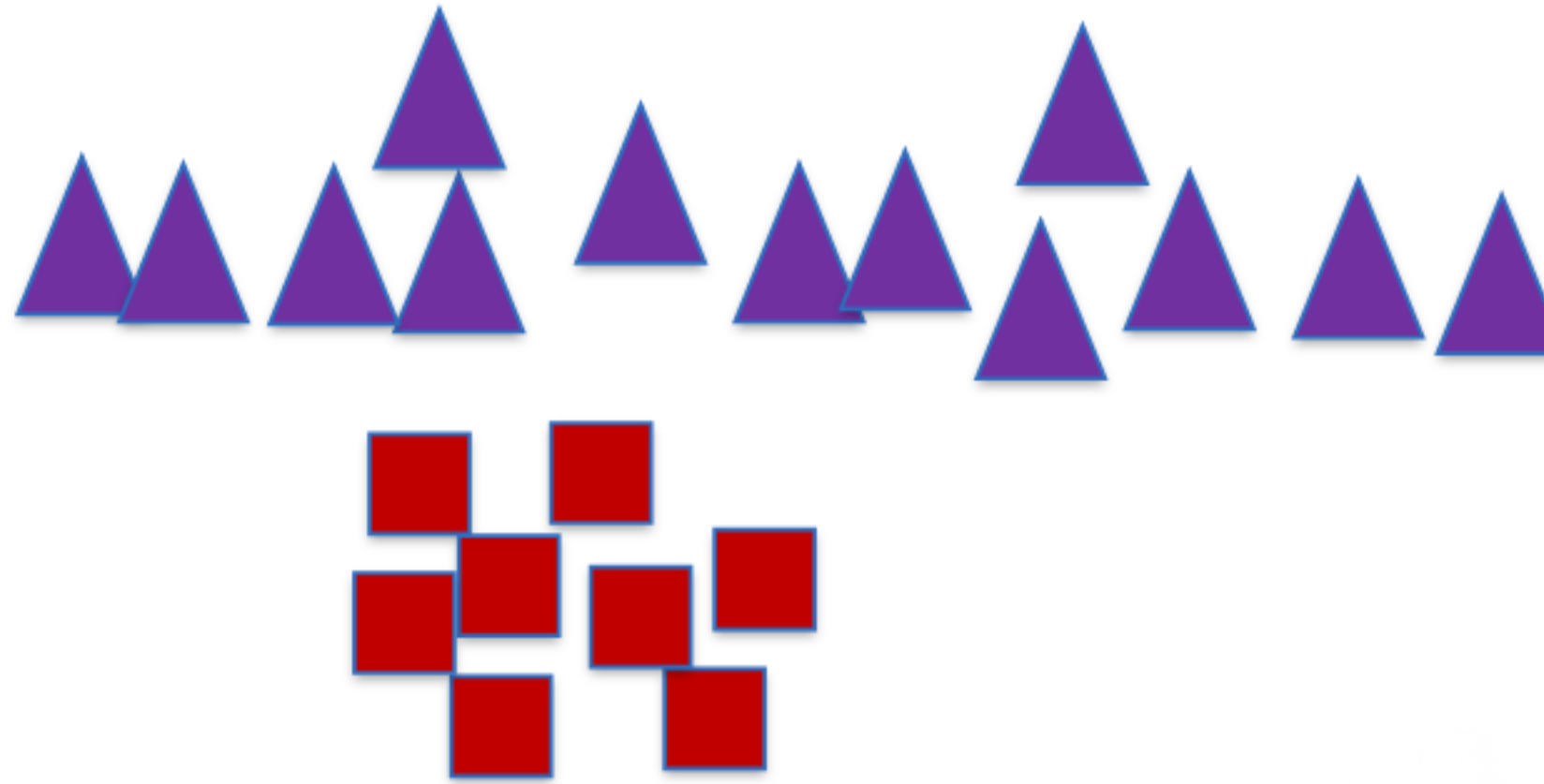
We cannot fit a line that will separate these two classes in this representation



Supervised Classification

- Solution: Transform to a space where they can be separated.

Apply a
transformation
 $Z = x^2 + y^2$



Decision Trees

- A very common method that is used to classify data is the decision tree
- The decision tree is easily interpretable, and essentially learns to split data based on values of features.
- Advantages: will ignore features that are unnecessary to the classification, when small they are easy to interpret, and have often been shown to be extremely good (esp. with random forests, bagging, etc.)

Decision Trees

- The decision tree is learned by using the concept of information entropy. Each split is chosen to maximize the purity of each daughter node of the tree.
- Numerous types of metrics can be used here, including KL divergence, or Gini Impurity, or Variance Reduction
- The main issue with decision trees is the tendency to overfit to the training data. Ensemble methods can reduce this tendency.

Unsupervised Classification

- In many cases, we do not have labeled examples. In this case, we can turn towards unsupervised methods
- The most common method is k-means clustering, but there are numerous others
- In the interests of time, we will simply explore some of the other clustering methods in a notebook

K-Means Clustering

- SVD, NMF and ICA are decompositions. But sometimes, we don't want to decompose our signal, but just group them into 'alike' sets.
- This is termed 'clustering'. The easiest and most widely used method is the k-means algorithm

K-means Clustering algorithm, to separate data $(x_1, x_2, \dots, x_n)$ into k clusters

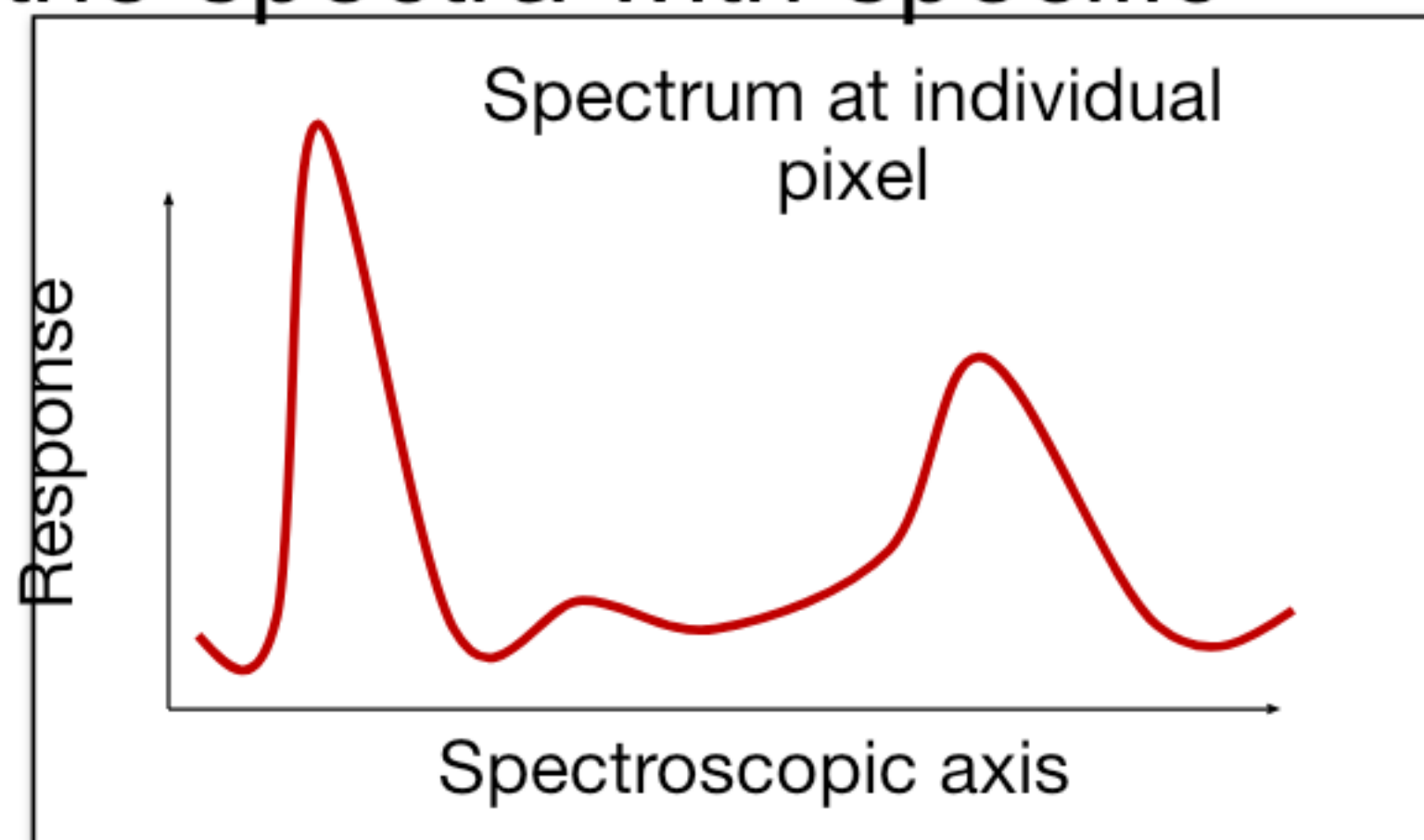
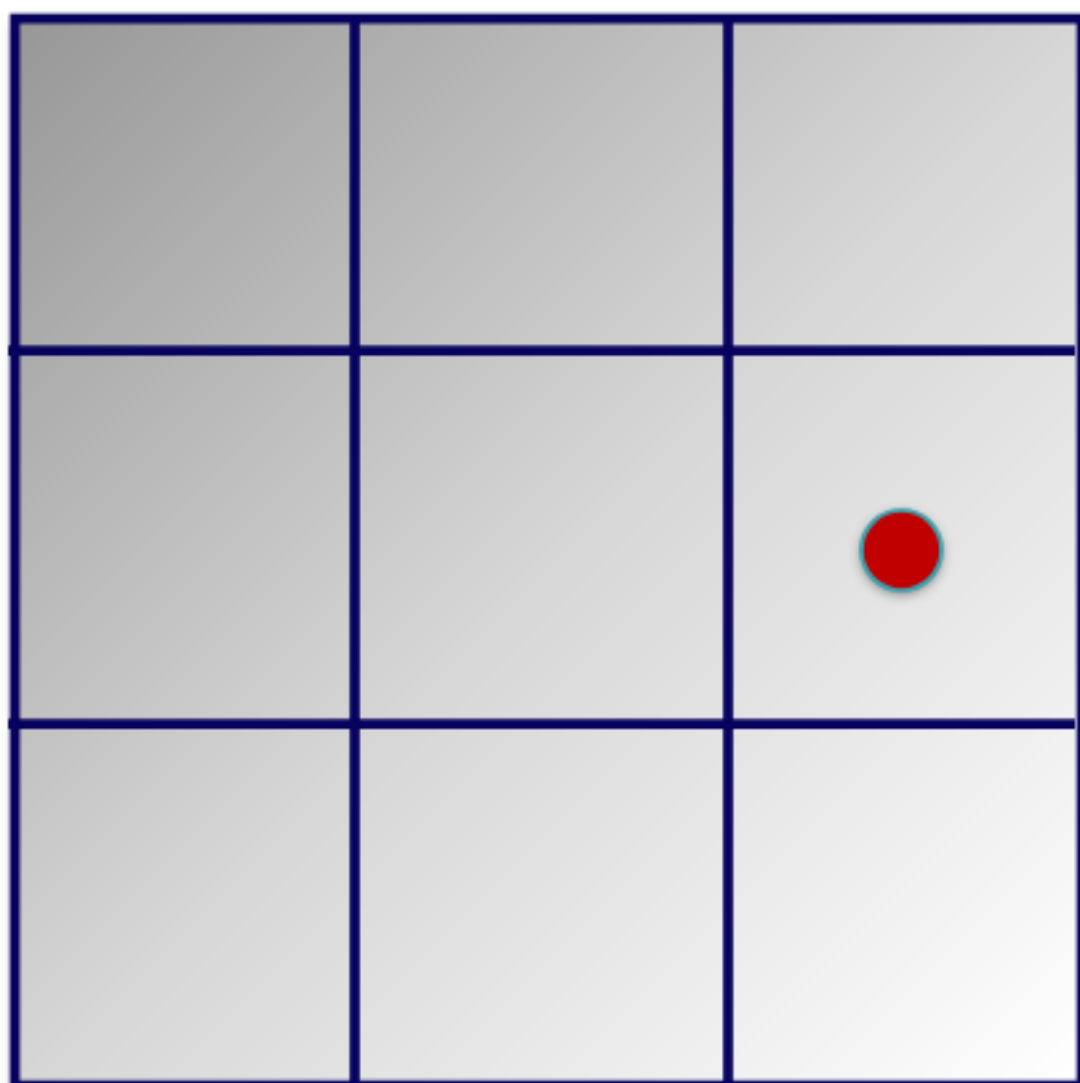
$$\arg \min_{\mathbf{S}} \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \quad \text{where } \boldsymbol{\mu}_i \text{ is the mean of points in } S_i$$

(Determine $\mathbf{S} = \{S_1, S_2, \dots, S_k\}$, such that within cluster sum of squares is minimized)

Spectral Data

- Many imaging modalities have spectroscopy components
- By capturing spectra across an image, it allows us to correlate features of the spectra with specific regions of the sample

Image (MxN pixels)



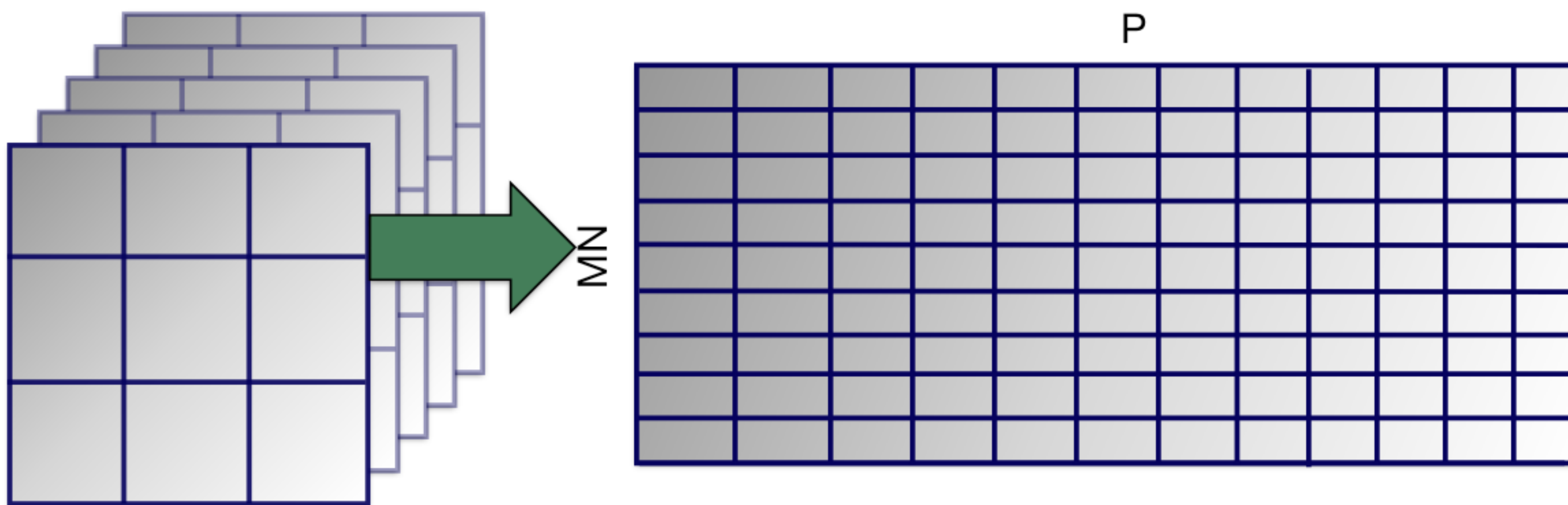
M (rows) x N (columns) x P (spectral points)

Spectral Data

- Examples include EELS, micro Raman imaging, force-distance curve maps in AFM, Scanning tunneling spectroscopy, etc.
- Usually, the 3D matrix is converted to a 2D matrix
 $M \text{ (rows)} \times N \text{ (columns)} \times P \text{ (spectral points)}$

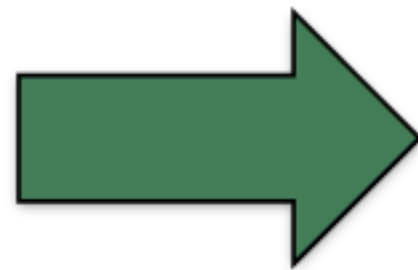
$$= (MN) \times P$$

Pixels x spectral points



Matrix Factorization

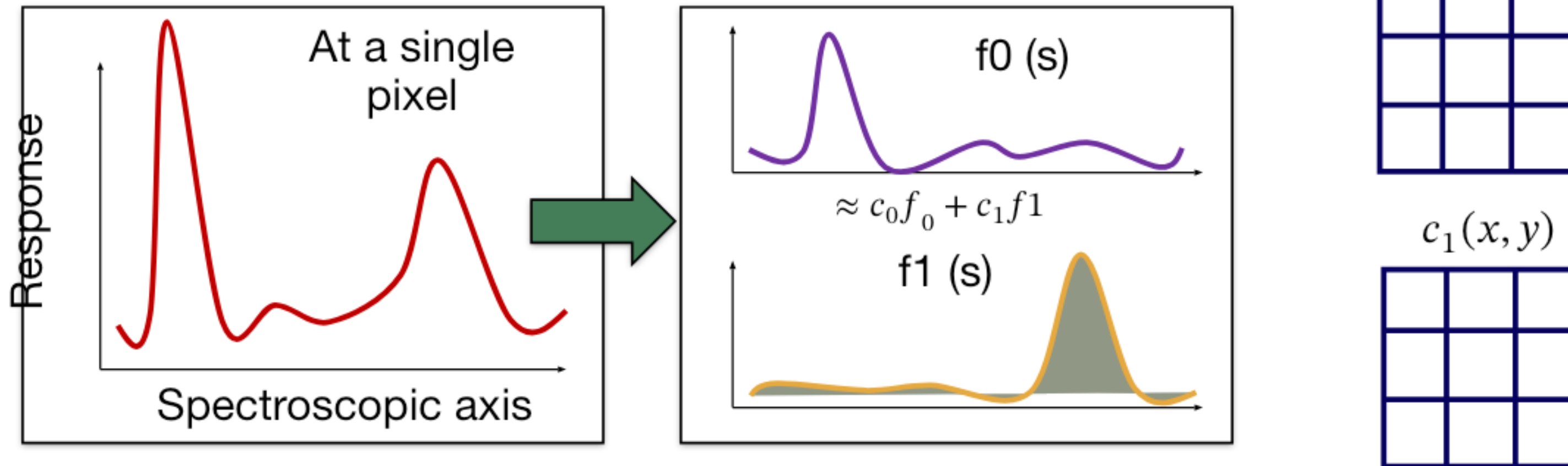
- We now have a 2D matrix (call it **X**) of size (MN,P)
- If **X** is 'small' (e.g., (10,10)) we can comprehend it easily. But if the size of X is (2500, 32), then this becomes more difficult
- The number of spectral points **P** is often termed the "number of dimensions".
- To see trends in our data, we want to visualize them in some manner. But the 3D data cube is big, so what can we do?



Matrix Factorization

Dimensionality Reduction

- The reason it is difficult to visualize is because there are too many spectral points P per pixel. What we need is to reduce the dimensionality, while preserving most of the data structure.
- Alternative viewpoint: Represent the spectrum at pixel z by a linear expansion



(1) Singular Value Decomposition

$$X = USVT$$

Note: In the full decomposition, $r = n$.
Practically, $r \ll n$

X is of size (m,n)
 U is of size (m,r)
 S is of size (r,r)
 V is of size (n,r)

The columns of U, V
form the basis
 S scales them, in
descending order

By the SVD theorem, it is always possible to decompose real matrix X into $USVT$ where

U, S, V : **unique**

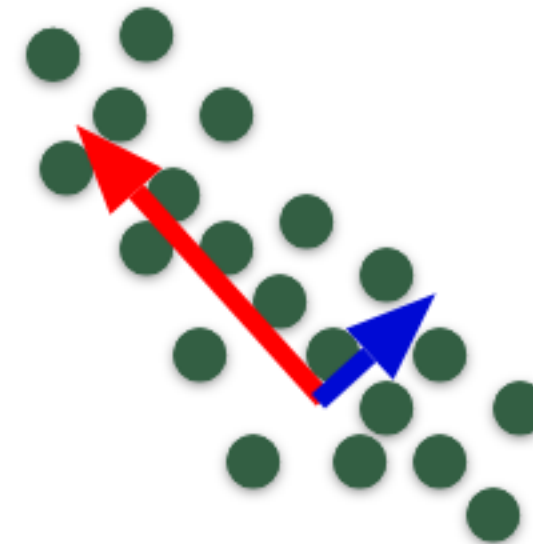
U, V : **column orthonormal**

- $UTU = I, VTV = I$
- Columns are orthogonal unit vectors

S : **diagonal**

- Positive, sorted in descending order

Finding the main
'directions' in the data



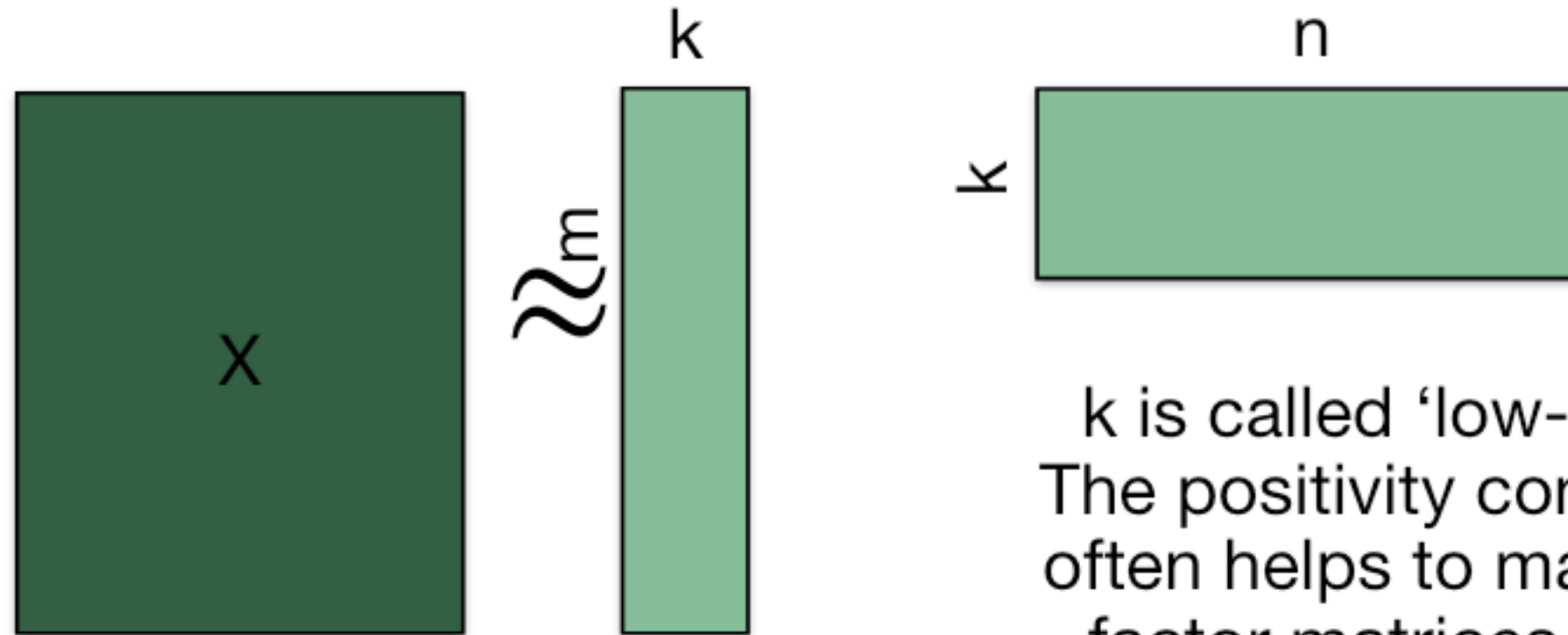
Non-negative Matrix Factorization

- SVD is an extremely powerful tool to visualize data and observe trends, but it has weaknesses. Mainly, it often produces highly unphysical behavior, for instance negative intensity values in spectra where only positive values are physically realizable
- Therefore, we can turn to other methods, such as non-negative matrix factorization approaches

Matrix Form

$$\mathbf{X} \approx \mathbf{WH}$$

Here, we enforce that X , W and H are all non-negative matrices



k is called 'low-rank'. The positivity constraint often helps to make the factor matrices more interpretable.

Spectral Unmixing: N-FINDR

- Spectra for a given pixel is assumed to be a linear combination of the end-member spectra (+ Gaussian noise). The mixing proportions sum to 1

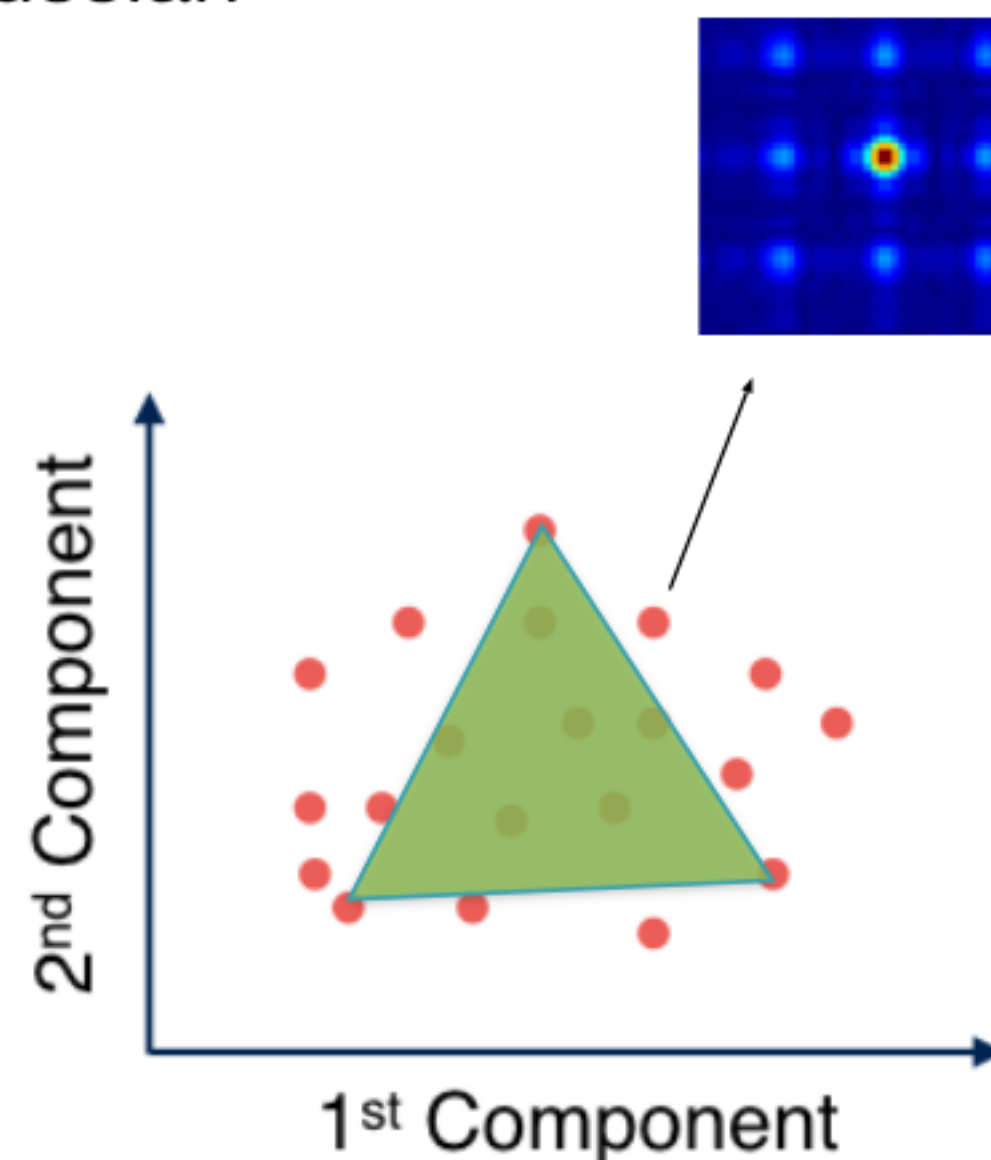
Physics constraint

$$p_{ij} = \sum_k e_{ik} c_{kj} + \varepsilon \quad \sum_k c_{kj} = 1$$

- Let E be the matrix of end-members (here, 3).

$$E = \begin{bmatrix} \vec{1} & \vec{1} & \vec{1} \\ e_1 & e_2 & e_3 \end{bmatrix} \quad V \left(\frac{1}{(l-1)!} \right) |\det(E)|$$

- Iteratively select endmembers, accepting the new selection if the volume increases



Segmenting based on phases

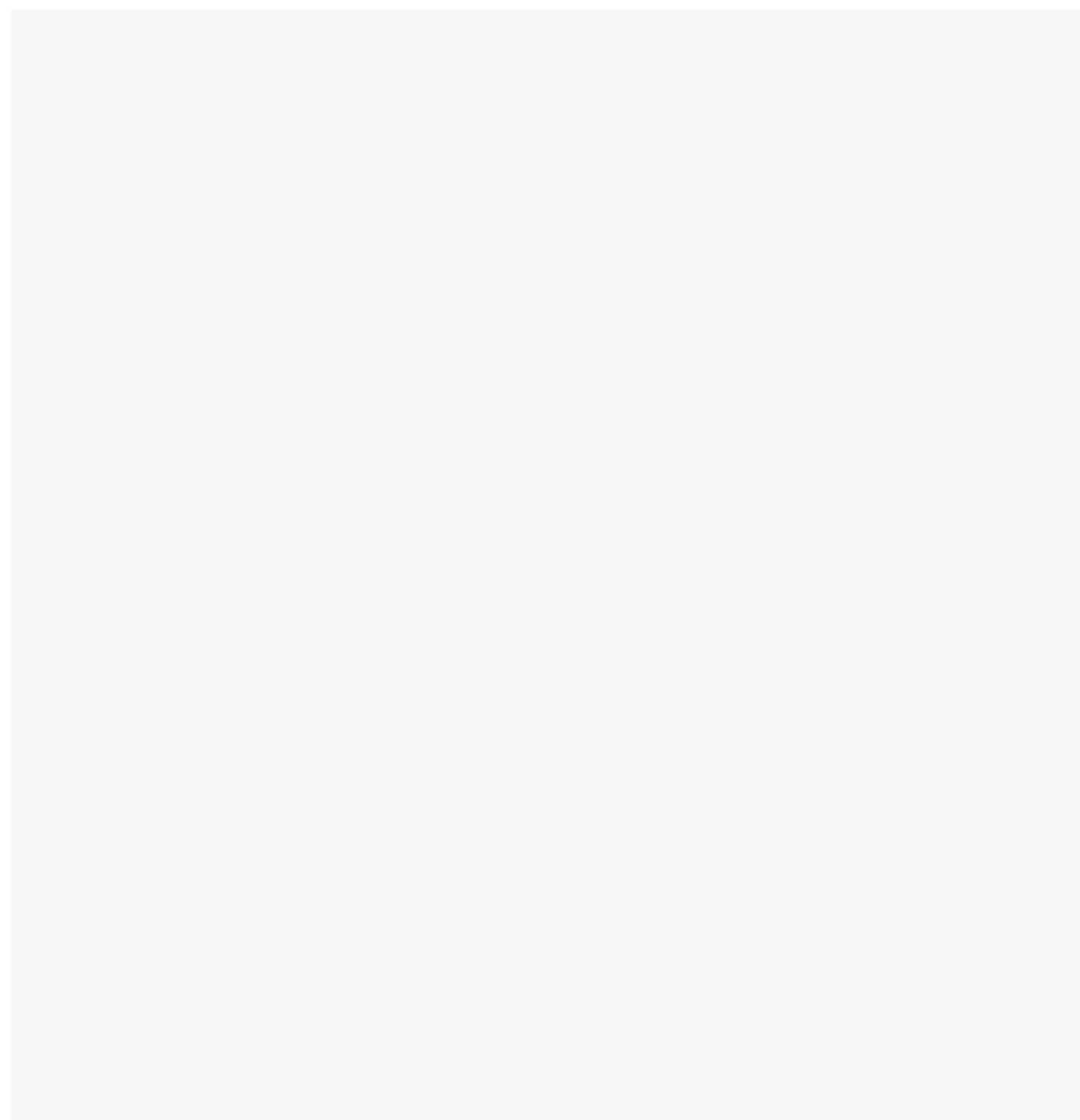
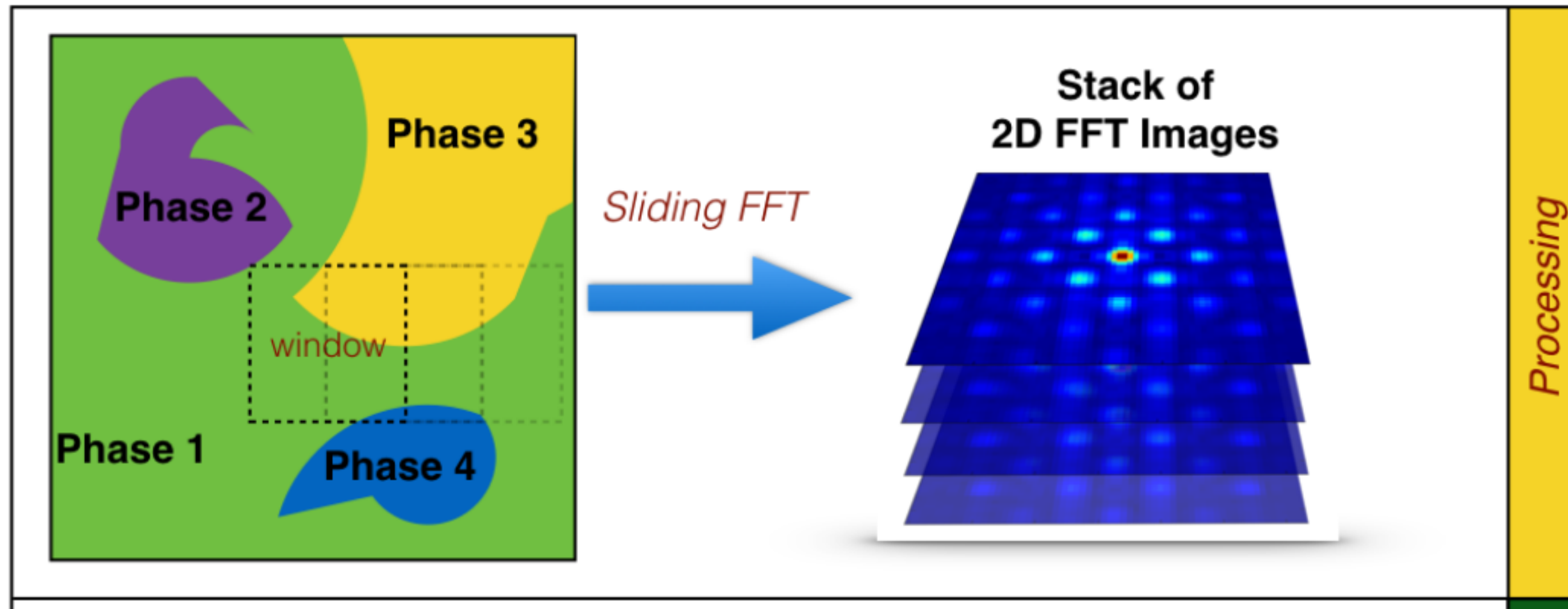


Image courtesy of
A. Borisevich (ORNL)
and
Q. He
(ORNL/Manchester)

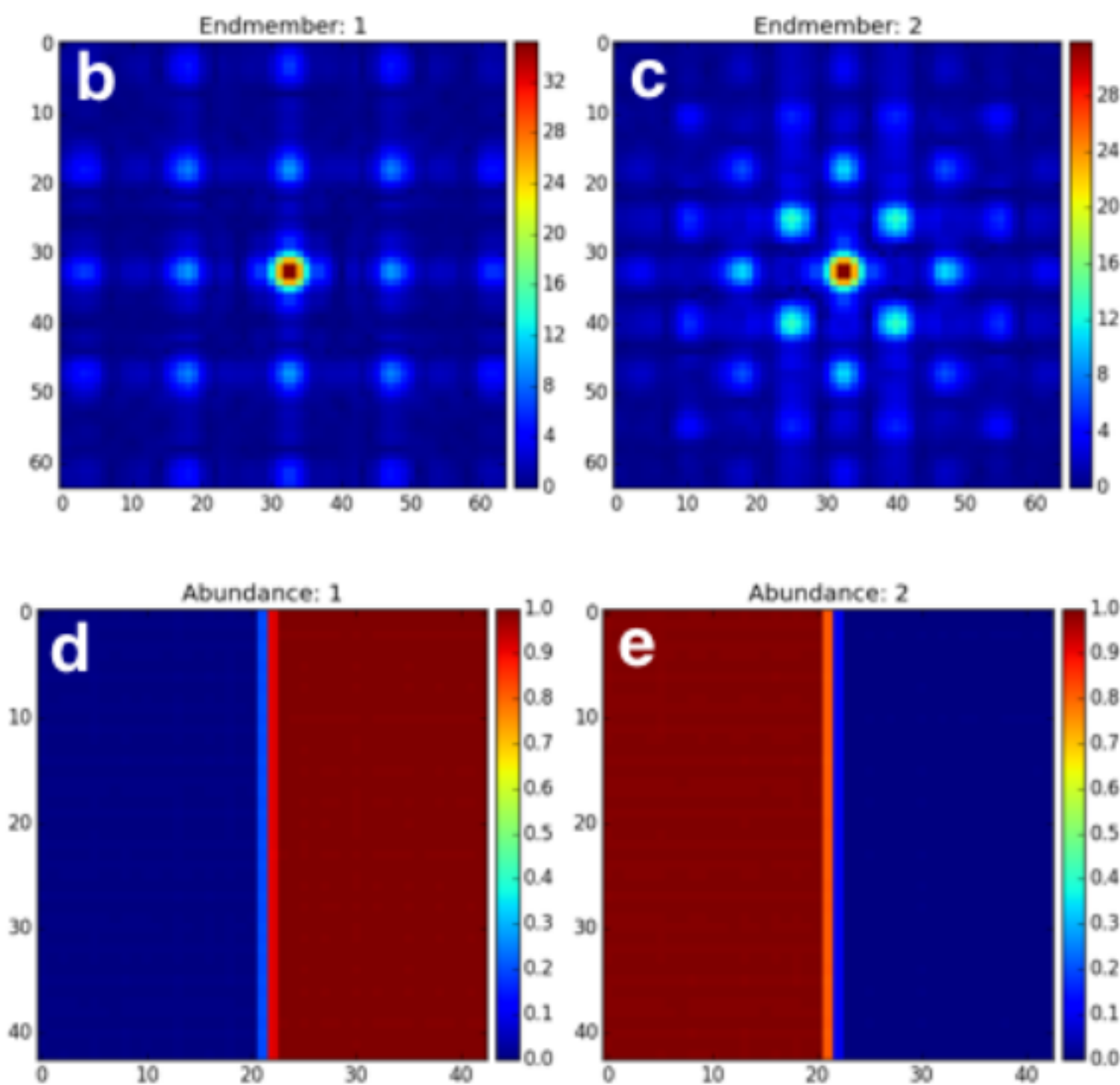
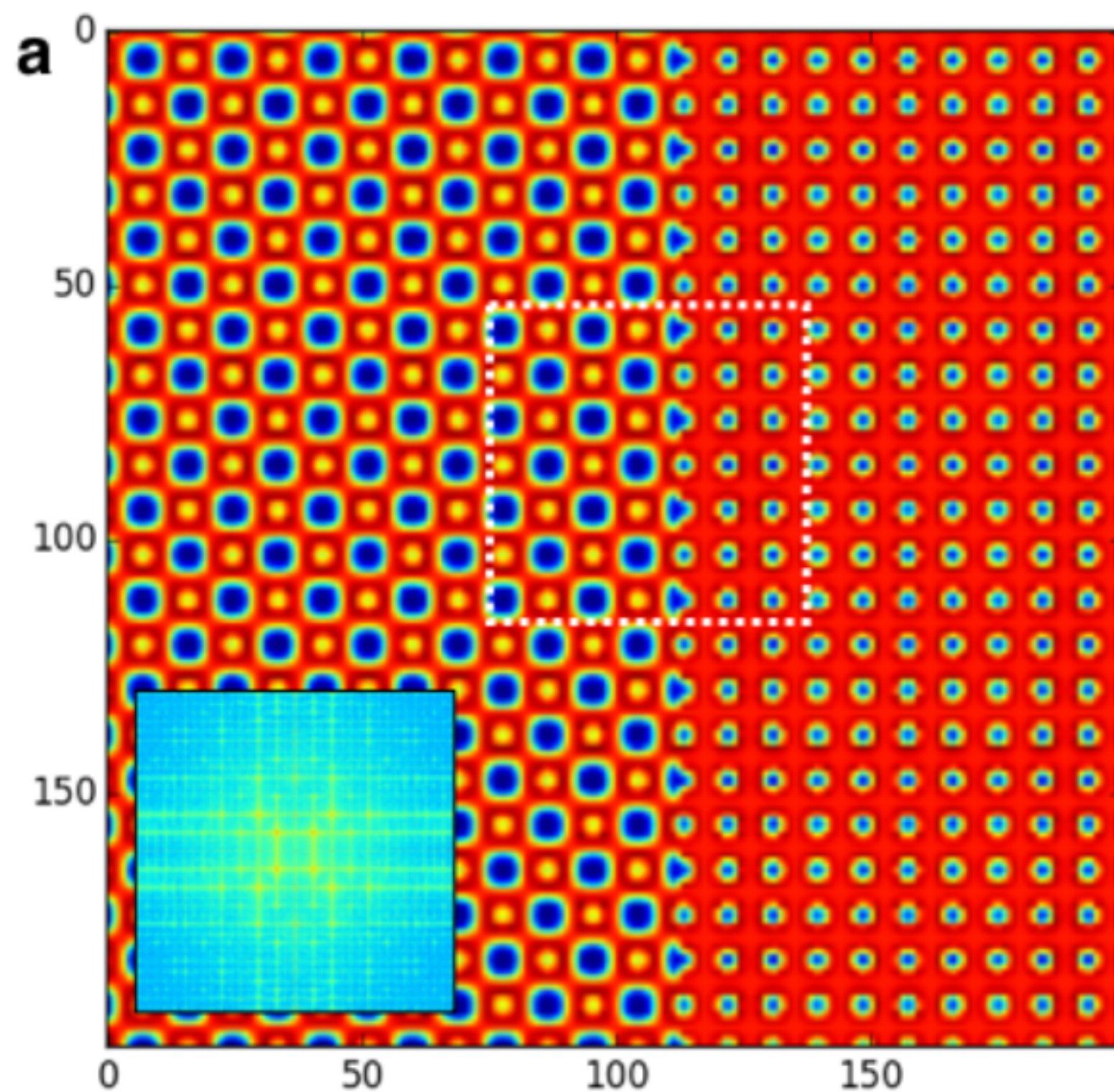
How can we segment this image based on the
phases present?

Sliding Window Method



Now, we can try to do spectral unmixing on the 2D stack of FFT images

Ideal test case



Spectral Unmixing: Notebook

Let's try out various unmixing methods including N-FINDR on the STEM image shown below

Open the notebook on spectral unmixing

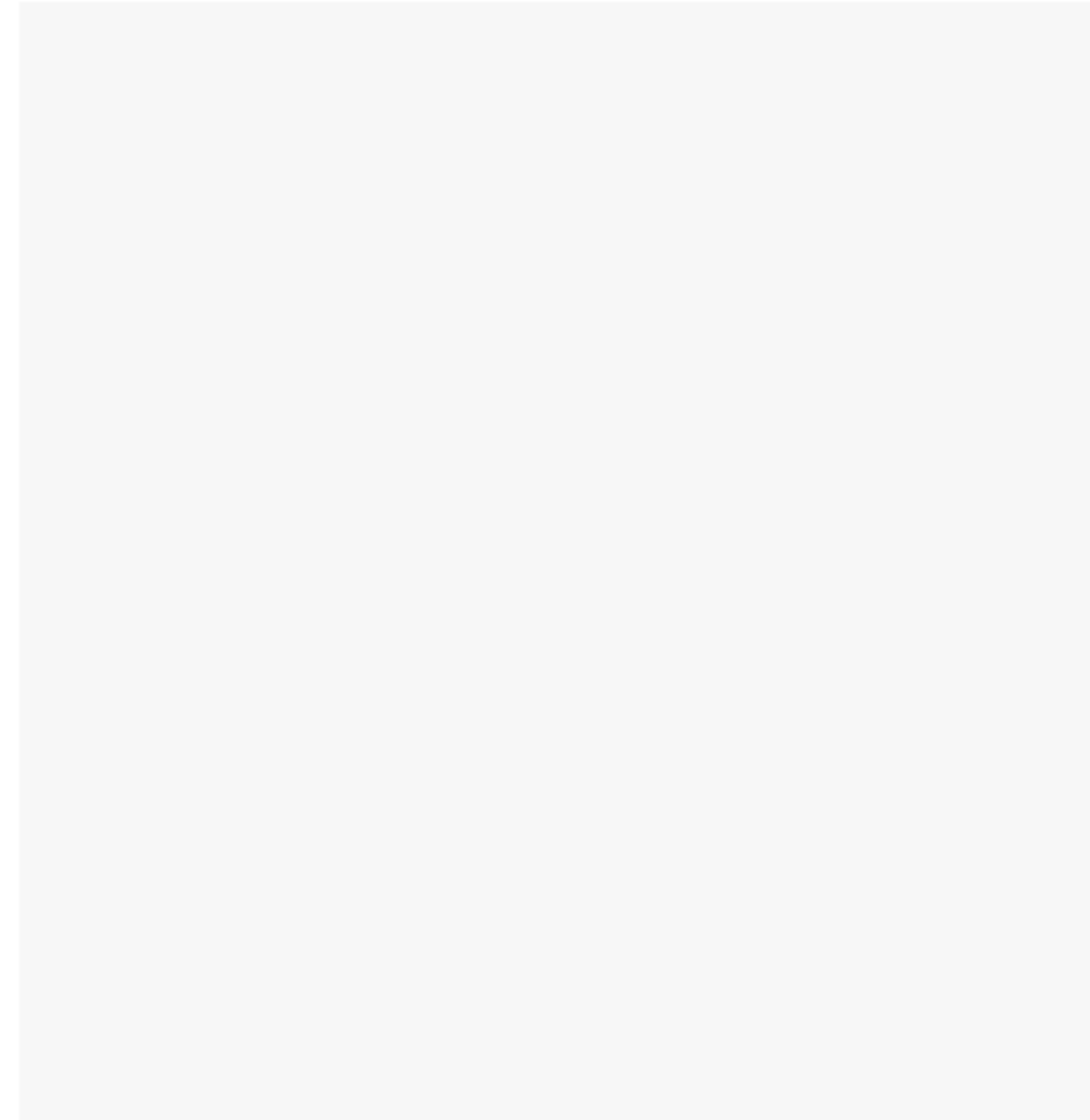


Image courtesy of
A. Borisevich (ORNL)
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An LLM is a next-token model wrapped in context

What the model learns

Patterns in language, code, tables, figures, and scientific text from training data. The trained weights are fixed during an ordinary chat.

What the prompt supplies

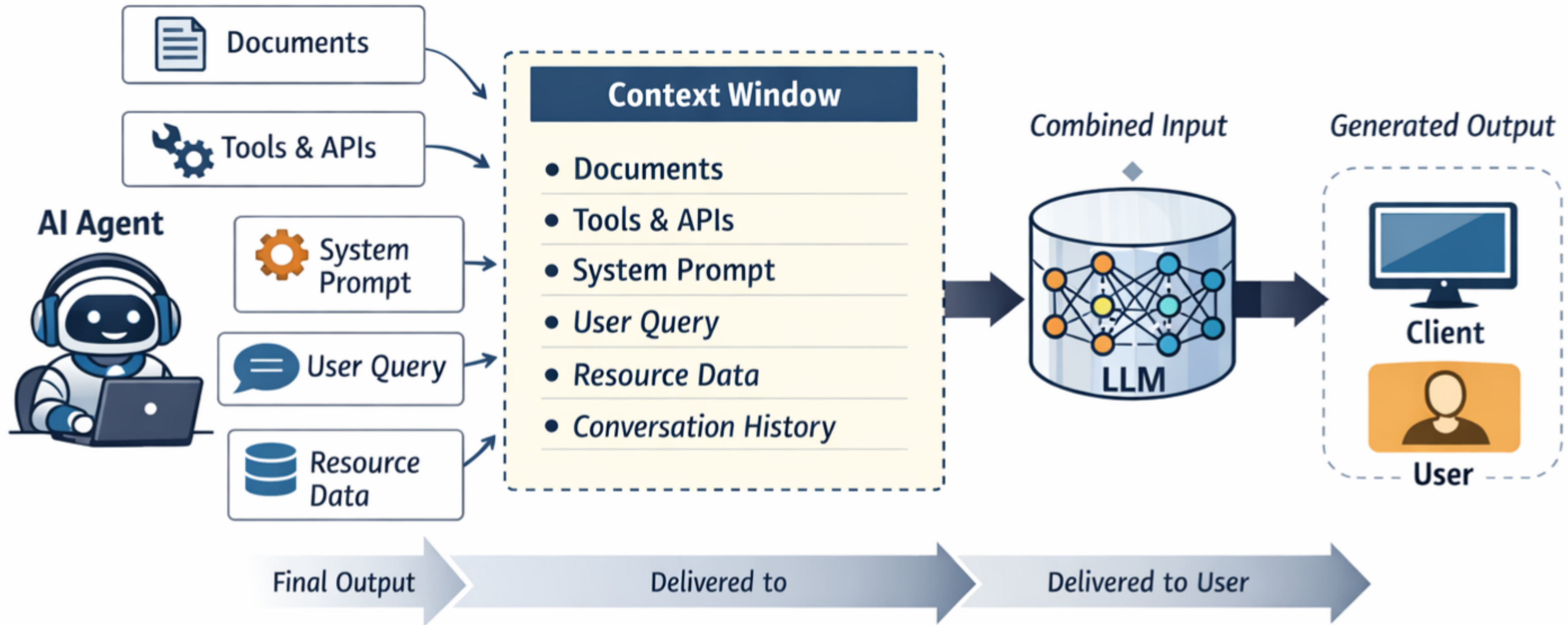
The current task, conversation history, retrieved documents, notebook state, and tool results. This is the working memory for the next response.

What makes it an agent

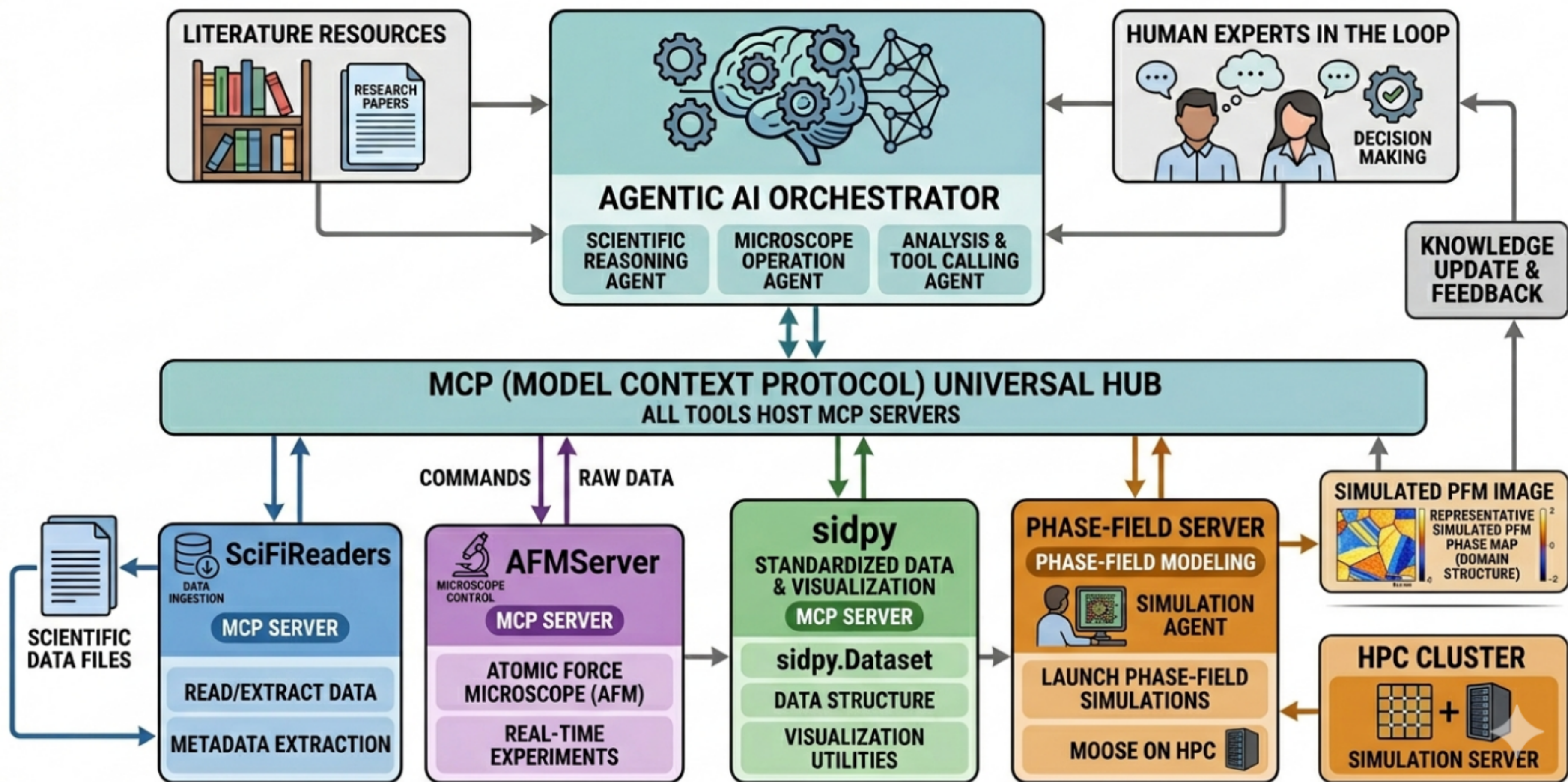
A loop that lets the LLM choose actions, call tools, observe results, and update the next step. In microscopy, those tools can include simulators, analysis code, and instrument APIs.



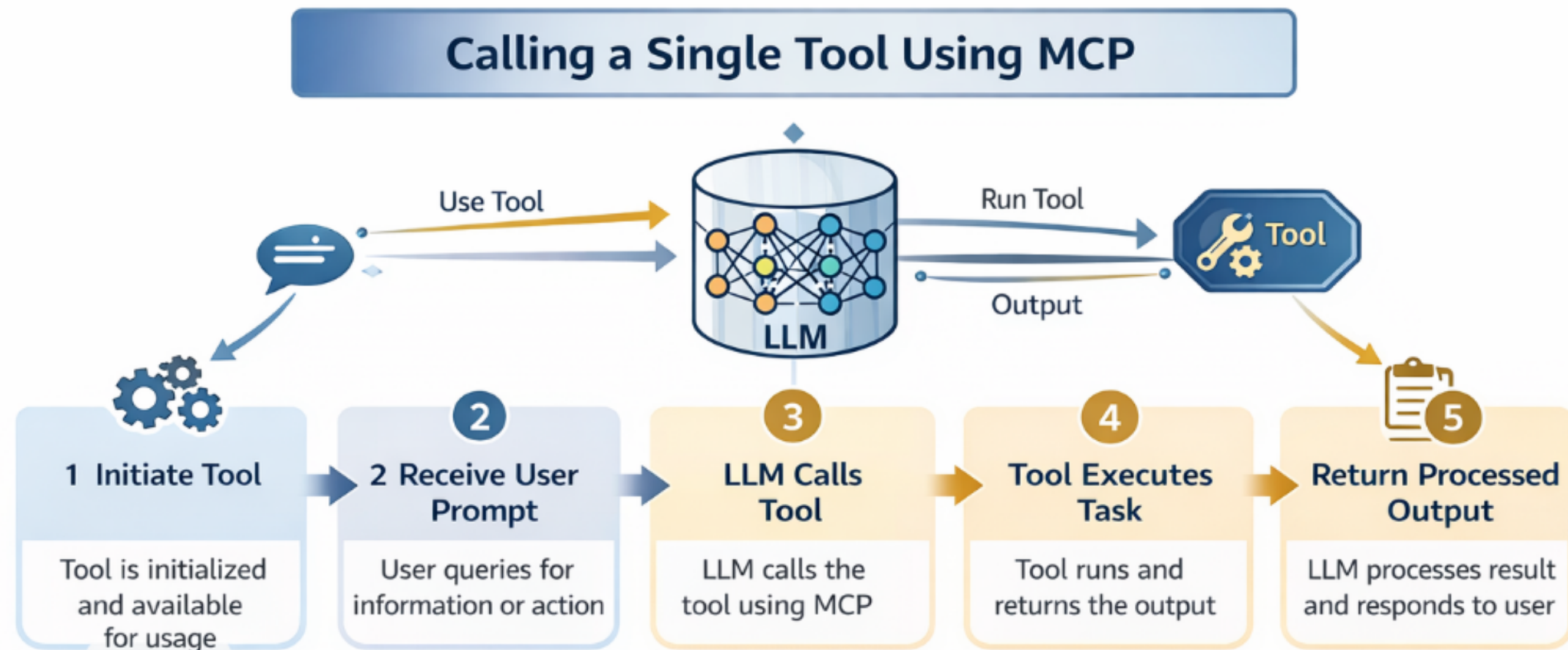
LLMs are stateless models



AI-DRIVEN MICROSCOPY WORKFLOW UTILIZING MCP SERVER ECOSYSTEM



MCP lets an LLM call tools, observe results, and continue



Specifics of MCP Tool Usage

Calling a Single Tool Using MCP

